

Class 2: Vector Equations

For now we will define a vector as an ordered n-tuple of real numbers. Usually we will write vectors as a column of numbers.

The set of all n-tuples is denoted by _____

Two operations on any set of vectors are

1.

2.

Linear Combinations

If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are vectors and $c_1, c_2, \dots, c_k$ are scalars, the expression _____

is called a linear combination of the $\mathbf{v}$'s.

Example

Key Idea Any system of linear equations can be written as a single vector equation.

Definition: The set of all linear combinations of the vectors $\{v_1, v_2, \dots, v_k\}$ is called

A vector, $\vec{b}$, is in the $\text{Span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ if

Key Idea: To decide if a given vector $\vec{b}$ is in the span of a set of vectors $v_1, \dots, v_k$ is the same as

Which is the same as

Finding a solution to the system is the same as finding a set of weights to express $\vec{b}$ as a linear combination.

Examples

Is the vector $\vec{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ in the span of $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$?

Chemical Production Problem

A chemical company operates three production facilities where three different solvents are made. The facilities are set up to produce different amounts of each solvent. The table below shows the number of gallons of each solvent produced from a single 50 gallon drum of raw material at each facility.

	Facility 1	Facility 2	Facility 3
Solvent 1	20	4	4
Solvent 2	10	14	5
Solvent 3	5	5	12

The company estimates that the demand is for 500 gallons of the first solvent, 850 gallons of second solvent and 1000 gallons of the third. What is the least number of 50 gallon drums of raw material that should be shipped to each facility to meet this demand?